

```
one@one:~$ cd /usr/local/apache-opennlp-1.5.3/bin/
one@one:/usr/local/apache-opennlp-1.5.3/bin$ gcc lib_inp_sentence.txt
(gcc) 15-11-2016 11:05:00
/usr/local/apache-opennlp-1.5.3/bin$ ./lib_inp_sentence.txt
How are you? I am Fine. What is your hobby? I
like music, reading books. Where are you now? I
am in India. I like Singapore, China. I would
like to go to France and Germany. Tomorrow
morning I will reach Sri Lanka. I am going to
watch movies of Bruce Lee and Jackie Chan with
Bruce Willis. I had good lunch by Yesterday
afternoon. I like Bill Gates.
```

Figure 1: Input text file

```
one@one:~$ cd /usr/local/apache-opennlp-1.5.3/bin/
one@one:/usr/local/apache-opennlp-1.5.3/bin$ ./opennlp SentenceTokenizer -t inp_sentence.t
xt
How are you? I am Fine. What is your hobby? I like music, reading books. Where are you now? I
am in India. I like Singapore, China. I would like to go to France and Germany. Tomorrow
morning I will reach Sri Lanka. I am going to watch movies of Bruce Lee and Jackie Chan with
Bruce Willis. I had good lunch by Yesterday afternoon. I like Bill Gates.
Average: 83.3 sent/s
Total: 1 sent
Runtime: 0.002s
one@one:/usr/local/apache-opennlp-1.5.3/bin$
```

Figure 2: Token separation

```
one@one:~$ cd /usr/local/apache-opennlp-1.5.3/bin/
one@one:/usr/local/apache-opennlp-1.5.3/bin$ ./opennlp SentenceDetector -t inp_sentence.t
xt
Loading Sentence Detector model ... done (0.355s)
How are you?
I am Fine.
What is your hobby?
I like music, reading books.
Where are you now?
I am in India.
I like Singapore, China.
I would like to go to France and Germany.
Tomorrow morning I will reach Sri Lanka.
I am going to watch movies of Bruce Lee and Jackie Chan with Bruce Willis.
I had good lunch by Yesterday afternoon.
I like Bill Gates.
Average: 785.9 sent/s
Total: 12 sent
Runtime: 0.017s
one@one:/usr/local/apache-opennlp-1.5.3/bin$
```

Figure 3: Sentence detection

```
one@one:~$ cd /usr/local/apache-opennlp-1.5.3/bin/
one@one:/usr/local/apache-opennlp-1.5.3/bin$ ./opennlp TokenizerME -t inp_sentence.t
xt
Loading Tokenizer model ... Loading Token Name Finder model ... done (0.557s)
How are you?
I am Fine.
What is your hobby?
I like music, reading books.
Where are you now?
I am in India.
I like Singapore, China.
I would like to go to France and Germany.
Tomorrow morning I will reach Sri Lanka.
I am going to watch movies of Bruce Lee and Jackie Chan with Bruce Willis.
I had good lunch by Yesterday afternoon.
I like Bill Gates.
Average: 25.6 sent/s
Total: 1 sent
Runtime: 0.039s
one@one:/usr/local/apache-opennlp-1.5.3/bin$
```

Figure 4: Finding the location of a name

`inp_sentence.txt` is shown in Figure 2.

Sentence detection

The sentence detector detects each sentence based on the punctuation symbols like the full stop (.) and question mark (?).

The result of the SentenceDetector tool using the pre-trained model `en-sent.bin` is shown in Figure 3.

Name finder (finding the location of a name)

The name finder can detect named entities and numbers in text – like the names of famous persons, locations and organisations. We need to have pre-trained models to detect entities. The example in Figure 4 shows the process of finding the location in a file `inp_sentence.txt` by using the model `en-ner-location.bin`. The locations are marked as `<START:location> India <END>`. The command given

```
one@one:~$ cd /usr/local/apache-opennlp-1.5.3/bin/
one@one:/usr/local/apache-opennlp-1.5.3/bin$ ./opennlp TokenizerME -t inp_sentence.t
xt
Loading Tokenizer model ... Loading Token Name Finder model ... done (0.536s)
How are you?
I am Fine.
What is your hobby?
I like music, reading books.
Where are you now?
I am in India.
I like Singapore, China.
I would like to go to France and Germany.
Tomorrow morning I will reach Sri Lanka.
I am going to watch movies of Bruce Lee and Jackie Chan with Bruce Willis.
I had good lunch by Yesterday afternoon.
I like Bill Gates.
Average: 10.2 sent/s
Total: 1 sent
Runtime: 0.009s
one@one:/usr/local/apache-opennlp-1.5.3/bin$
```

Figure 5: Finding a person

```
inp_sentence.txt
I would like to work in Microsoft in USA. But I
am currently working at Dell for doing
collaborative work with IBM to Share the
knowledge of Intel.
```

Figure 6: The input file for finding an organisation

```
one@one:~$ cd /usr/local/apache-opennlp-1.5.3/bin/
one@one:/usr/local/apache-opennlp-1.5.3/bin$ ./opennlp TokenizerME -t inp_sentence.t
xt
Loading Tokenizer model ... Loading Token Name Finder model ... done (0.557s)
How are you?
I am Fine.
What is your hobby?
I like music, reading books.
Where are you now?
I am in India.
I like Singapore, China.
I would like to go to France and Germany.
Tomorrow morning I will reach Sri Lanka.
I am going to watch movies of Bruce Lee and Jackie Chan with Bruce Willis.
I had good lunch by Yesterday afternoon.
I like Bill Gates.
Average: 19.6 sent/s
Total: 1 sent
Runtime: 0.051s
done (0.478s)
I would like to work in <START:organization> Microsoft <END> in USA.
But I am currently working at Dell for doing collaborative work with <
START:organization> IBM <END> to Share the knowledge of <START:organi
zation> Intel <END>.
Average: 7.9 sent/s
Total: 1 sent
Runtime: 0.126s
one@one:/usr/local/apache-opennlp-1.5.3/bin$
```

Figure 7: The output when finding an organisation

in Figure 4 is actually the pipelining of both tokenising and token name finding using the tools `TokenizerME` and `TokenNameFinder`, respectively.

Name finder (finding a person)

The example given in Figure 5 shows the process of finding the person in a file `inp_sentence.txt` by using the model `en-ner-person.bin`. The persons are marked as `<START:person> Bill Gates <END>`. The command in Figure 5 actually pipelines both tokenising and token name finding, using the tools `TokenizerME` and `TokenNameFinder`, respectively.

Name finder (finding an organisation)

Figures 6 and 7 show the process of finding an organisation in a file `inp_sentence.txt` by using the model `en-ner-organization.bin`. The results are marked as `<START:organization> Intel <END>`. The command in Figure 6 is actually pipelining both tokenising and token name finding using the tools `TokenizerME` and `TokenNameFinder`, respectively.

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